

PALANI PRASHANTH B

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PROFESSIONAL SUMMARY

AI Researcher and Content Strategist with an M.Tech in Computer Science (Big Data Analytics), focused on applied AI systems, LLM-powered applications, and research-driven content. Hands-on builder of RAG pipelines, Agentic AI tools, and MCP-based systems, with experience translating complex AI concepts into clear technical documentation. Strong working knowledge of modern AI tooling for production workflows, content creation, and rapid prototyping.

RESEARCH & PROFESSIONAL EXPERIENCE

AI Researcher & AI Content Strategist (Remote) — AICE — AI Center of Excellence

Nov 2025 – Present | Full-time

- Research and evaluate emerging AI models, frameworks, and tools (LLMs, RAG, Agentic AI, MCP) to identify what is genuinely production-ready vs experimental — and translate findings into working prototypes.
- Built 5 production-grade applied AI applications spanning RAG systems, MCP servers, code-analysis tools, and LLM playgrounds, validating which architectures hold up under real workloads (see AICE Project Portfolio below).
- Author technical content and structured learning material — explainers, deep-dives, and tutorials on LLMs, RAG, Agentic AI, evaluation methods, and prompt engineering — aimed at applied practitioners.
- Manage AICE's daily AI updates social media presence across Instagram, LinkedIn, Facebook, X, and YouTube — using ChatGPT for content creation, Gemini and Claude for carousel preparation, and Perplexity / Comet for engagement and growth analysis.
- Collaborate with researchers and developers to translate complex AI research into clear technical write-ups, reports, and reference material.
- Use modern AI tooling (Claude, ChatGPT, Gemini, Gamma, Canva, etc.) extensively in production workflows for content creation, research synthesis, and rapid prototyping.

Video Annotation Specialist (Remote) — Nexority

Oct 2025 – Nov 2025 | Contract

- Annotated large-scale video and vision datasets for AI/ML training pipelines, following strict guidelines and ensuring inter-annotator consistency.
- Conducted quality assurance (QA) audits and supported model evaluation by identifying annotation gaps, edge cases, and data quality issues.
- Gained hands-on exposure to real-world data collection challenges, dataset bias awareness, and performance validation.

TECHNICAL SKILLS

Programming & AI/ML: Python, JavaScript/TypeScript, Node.js, Machine Learning, Deep Learning, NLP, Computer Vision, LangChain, T5, YOLO, OpenCV, TensorFlow

LLMs & Frameworks: Llama 3 (70B), Mixtral, Gemma via Groq; HuggingFace embeddings, ChromaDB, Qdrant, MCP (Model Context Protocol), FastAPI, Streamlit, Gradio

Data & Analytics: SQL (MySQL), MongoDB, BigQuery, Tableau, Matplotlib, ELK Stack (Elasticsearch, Logstash, Kibana)

Cloud & Infra: AWS, Google Cloud Platform (GCP), Microsoft Azure, Cloudflare Tunnel, Razorpay (payments)

Frontend: React, Vite, Tailwind CSS, Lucide, React Markdown

Tools & Platforms: Git/GitHub, Arduino, CVAT (annotation)

Research & Evaluation: Data annotation pipelines, benchmarking concepts, QA audits, bias & fairness awareness, model evaluation

AI TOOLS & PRODUCTIVITY (Used in production at AICE)

LLM Chat & Coding: Claude (chat, Cowork, Claude Code), ChatGPT, Gemini, Grok, Cursor, Codex, Lovable, Bolt, Antigravity

Agents & Automation: Zapier Agents, n8n; research with Perplexity, Comet, ChatGPT Atlas (social-growth analysis)

Content & Media: Gamma, Canva, Beautiful.ai, Claude Design (course decks); ElevenLabs, Suno, Sarvam (audio); HeyGen, Higgsfield, Google Veo 3 (video); ChatGPT Image, Gemini Nano Banana (image)

Knowledge Management: Notion (knowledge bases and AI grounding)

AICE PROJECT PORTFOLIO — 5 Production AI Applications

Built and shipped during tenure at AICE; each application explores a different production AI architecture.

SkySense AI — Agentic Flight Search & Booking Platform [\[GitHub\]](#)

MCP, Node.js, TypeScript, React, Tailwind, Groq (Llama 3), Amadeus, SerpAPI, Razorpay

Built an end-to-end agentic flight assistant: started as an MCP server exposing flight search to Claude Desktop and other LLM clients, then extended into a full-stack React + Node app with multi-provider search (Amadeus + Google Flights via SerpAPI) and Groq-based natural-language query parsing.

Implemented seat-map selection and Razorpay payment flow with strict tool-use guardrails to prevent LLM hallucination of seat numbers or payment details.

Git-Sense — AI-Powered Repository Analyzer [\[GitHub\]](#)

FastAPI, React, ChromaDB, LangChain, Groq

Built a RAG-based code analysis tool that ingests any public GitHub repository, indexes it for semantic search, and runs automated audits — bug/security risk detection, refactoring suggestions, architecture summaries, and auto-generated READMEs.

Added a "Push to Git" workflow to commit generated artifacts back to the source repo, plus persistent caching for instant repeat analyses.

DocSense — Local-First RAG Document Chat [\[GitHub\]](#)

React 19, Vite, Tailwind v4, FastAPI, LangChain, Qdrant, Groq (Llama 3 70B / Mixtral)

Developed a private, local-first RAG application that lets users chat with PDF/DOCX/TXT documents while keeping document storage entirely on-device, backed by FastAPI and a Qdrant vector index.

Exposed multi-model switching across Llama 3.3 70B, Mixtral, and other Groq models, plus auto-generated question suggestions to help users explore newly uploaded documents.

PromptSense — RAG Workbench with Evaluation Dashboard [\[GitHub\]](#)

React, FastAPI, Groq (Llama 3 70B), Python

Built a RAG workbench supporting URL and file ingestion, real-time streaming answers, and a Compare Mode for A/B testing different prompt strategies side-by-side.

Shipped an evaluation dashboard tracking cost, latency, and user feedback, plus auto-generated quick-questions per uploaded document.

ModelSense — LLM Playground & Prompt-Engineering Lab [\[GitHub\]](#)

React, Vite, Tailwind, Groq API

Designed a "Mission Control" UI plus a Mini Lab for prompt engineering — real-time control over Temperature, Top-P, system prompts, and model selection (Llama 3, Mixtral) with multi-persona chat presets.

Implemented JSON-mode enforcement for developer workflows and secure browser-local API key storage.

APPLIED AI RESEARCH

MCP Connector Research — Swiggy & Zomato in Claude — *MCP, Claude, Food-delivery APIs*

- Researched how MCP (Model Context Protocol) connectors operate inside Claude — connection lifecycle, tool discovery, authentication, and how natural-language requests get routed to third-party APIs.
- Used Swiggy and Zomato MCP connectors as case studies to document end-to-end flow (search → cart → checkout) and identify integration patterns useful for building production-grade connectors at AICE.

ACADEMIC PROJECTS

AI-Powered Welding Groove Analysis System — *YOLOv8-pose, OpenCV, CVAT, Gradio*

- Developed a computer-vision system using YOLOv8-pose keypoint detection to analyze U- and V-shaped welding grooves.
- Annotated image and video datasets in CVAT with QA checks; computed weld metrics and filler volume; shipped a Gradio interface for visual analysis.

Agentic AI Question Paper Generation System — *Groq, LangChain, T5, spaCy*

- Built an agentic AI platform that generates MCQs from PDF source material, integrating Bloom's Taxonomy for cognitive-level alignment.
- Achieved 97% precision and 94% recall on generated questions; used spaCy-based knowledge graphs to improve question relevance.

AI-Based Missing Person Identification System — *TensorFlow, YOLOv8, OpenCV, Pandas, MySQL*

- Built a facial recognition pipeline matching static reference images against real-time video feeds with ~95% accuracy.
- Managed datasets and lookup workflows in MySQL with a Python/Pandas data layer.

EDUCATION

M.Tech — Computer Science Engineering (Big Data Analytics)

Vellore Institute of Technology, Vellore | 2023 – 2025 (Completed)

B.E — Computer Science Engineering

Rajalakshmi Engineering College, Chennai | 2019 – 2023

CERTIFICATIONS & WORKSHOPS

- AWS Academy Cloud Foundations (Certification)
- IBM Project: Virtual Eye — AI Drowning Detection
- UK-India Education and Research Initiative (UKIERI)
- Robotics Training Program — Texas Instruments Kit (Workshop)
- Advanced Drone Technology Workshop — India Space Lab

EVENTS & ACTIVITIES

AICE Sales Hackathon (Hyderabad) — Organizer

April 2026

- Co-organized AICE's Sales Hackathon in Hyderabad, including planning, on-ground coordination, and product launch logistics.
- Supported the official launch of AICE Education and AICE Services at the event.